

A Rigorous Foundation of Gauss–Hermite and Gauss–Legendre Quadrature with Applications to the Generalized Interpolation Lattice Boltzmann Method

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Abstract

We present a unified, rigorous theoretical framework for the Gauss–Hermite and Gauss–Legendre quadrature rules and elucidate their complementary roles within the Generalized Interpolation Lattice Boltzmann Method (GILBM). Working in weighted L^2 spaces, we recover the orthogonality and completeness structure of the classical orthogonal polynomial families [21] and provide a self-contained five-step proof of the Gaussian quadrature precision theorem, establishing algebraic exactness of degree $2n - 1$ for any n -point rule whose nodes coincide with the roots of an orthogonal polynomial of degree n [18, 4]. By carefully distinguishing three structural roles—the *target functional* $\langle f, 1 \rangle_w$, the *quadrature nodes* (polynomial roots), and the *orthogonality-induced annihilation mechanism*—we correct a common conceptual conflation in the applied literature.

In the kinetic-theoretic setting we prove that the Hermite weight function is the *unique* member of the classical orthogonal families that coincides with the dimensionless Maxwell–Boltzmann distribution; this isomorphism is the structural reason Gauss–Hermite quadrature constitutes the canonical discretization of velocity space in the lattice Boltzmann method [7, 8, 16, 17]. Conversely, the physical-space reconstruction step intrinsic to GILBM corresponds to an unweighted integration on the unit cell and is therefore naturally governed by Gauss–Legendre quadrature. We further show, as a consequence of the completeness-interval theorem, that the two quadrature schemes are *not interchangeable*: each is rigidly tied to the support of its own weight measure. The framework provides a rigorous foundation for the design of discrete velocity sets, interpolation stencils, and a priori error estimates in higher-order GILBM solvers.

Keywords: Gauss–Hermite quadrature; Gauss–Legendre quadrature; orthogonal polynomials; weighted L^2 spaces; lattice Boltzmann method; generalized interpolation; Maxwell–Boltzmann distribution.

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1. Introduction

The lattice Boltzmann method (LBM) has, since the seminal work of He and Luo [7], been elevated from a heuristic discrete-kinetic model into a rigorous Galerkin-type discretization of the continuous Boltzmann equation. The mathematical machinery underpinning this elevation rests on two pillars: (i) the Hermite-polynomial expansion of the equilibrium distribution about the local Maxwellian, and (ii) the Gauss–Hermite quadrature that exactly preserves a finite hierarchy of velocity moments [16, 13, 17].

In the Generalized Interpolation Lattice Boltzmann Method (GILBM), the requirement to reconstruct the post-collision distribution at off-lattice locations introduces a *second* quadrature problem, defined this time on a bounded local cell with unit weight. This second problem

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is naturally addressed by Gauss–Legendre quadrature [9, 11, 20]. The two quadratures, far from being interchangeable, occupy distinct mathematical niches dictated by the completeness intervals of their underlying orthogonal-polynomial families.

The present paper has three objectives:

- (i) **Mathematical:** to provide a self-contained, rigorous derivation of Gauss–Hermite and Gauss–Legendre quadrature within a unified weighted- L^2 framework, including the algebraic-precision theorem with a complete proof and an explicit optimality argument.
- (ii) **Conceptual:** to dispel a common misconception that the Gaussian quadrature integral represents an inner product against the orthogonal polynomial ϕ_n . We show that the *target* functional is $\langle f, 1 \rangle_w$ and that ϕ_n acts only through the *node placement* and the *orthogonality annihilation* mechanism.
- (iii) **Applied:** to formulate the canonical assignment of Gauss–Hermite to velocity space and Gauss–Legendre to physical space within GILBM, and to prove (via the completeness-interval theorem) that this assignment is unique up to trivial rescaling.

2. Mathematical Preliminaries

2.1 Weighted L^2 spaces

Let $I \subseteq \mathbb{R}$ be an open interval (possibly unbounded) and $w: I \rightarrow \mathbb{R}_{\geq 0}$ a measurable weight function satisfying

- (W1) $w(x) \geq 0$ almost everywhere on I ;
- (W2) $\int_K w(x) dx < \infty$ for every compact $K \subset I$;
- (W3) all moments are finite: $\mu_k := \int_I x^k w(x) dx < \infty$ for all $k \in \mathbb{N}_0$.

The weighted Hilbert space

$$L^2(I, w) = \left\{ f: I \rightarrow \mathbb{R} \mid \int_I |f(x)|^2 w(x) dx < \infty \right\}$$

is endowed with the inner product and norm

$$\langle f, g \rangle_w := \int_I f(x) g(x) w(x) dx, \quad \|f\|_w := \sqrt{\langle f, f \rangle_w}.$$

2.2 Classical orthogonal polynomials

Applying Gram–Schmidt orthogonalization to $\{1, x, x^2, \dots\}$ inside $L^2(I, w)$ produces a family $\{\phi_n\}_{n=0}^\infty$ of polynomials with $\deg \phi_n = n$ and

$$\langle \phi_n, \phi_m \rangle_w = h_n \delta_{nm}, \quad h_n > 0.$$

The four classical families relevant here are summarized in Table 1.

Theorem 2.1 (Completeness; [21], Thm. 5.7.1). *If (I, w) satisfies (W1)–(W3) together with the determinacy of the Hamburger moment problem, then $\{\phi_n\}_{n=0}^\infty$ is dense in $L^2(I, w)$, i.e. it constitutes a complete orthogonal basis.*

Both Legendre and Hermite families satisfy the determinacy hypothesis. For Hermite this follows from Carleman’s criterion since $\sum_n \mu_{2n}^{-1/(2n)} = +\infty$.

Remark 2.2 (Completeness interval). The interval I on which completeness is asserted is *intrinsic* to the polynomial family. Any quadrature derived from $\{\phi_n\}$ inherits this interval as its natural domain; departing from I destroys both orthogonality and the precision guarantees built upon it.

Table 1: Classical orthogonal polynomial families.

Family	I	$w(x)$	h_n
Legendre P_n	$[-1, 1]$	1	$\frac{2}{2n+1}$
Chebyshev (1st) T_n	$[-1, 1]$	$(1-x^2)^{-1/2}$	$\pi/2$ ($n \geq 1$), π ($n = 0$)
Laguerre L_n	$[0, \infty)$	e^{-x}	1
Hermite (probabilist) He_n	$\mathbb{R}$	$(2\pi)^{-1/2} e^{-x^2/2}$	$n!$
Hermite (physicist) H_n	$\mathbb{R}$	e^{-x^2}	$\sqrt{\pi} 2^n n!$

2.3 Legendre and Hermite polynomials

Legendre. Three-term recurrence:

$$P_0(x) = 1, \quad P_1(x) = x, \quad (n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Rodrigues' formula: $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2-1)^n]$. Generating function [22]:

$$G_P(x, t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad |t| < 1.$$

Hermite (probabilist). Three-term recurrence:

$$\text{He}_0(x) = 1, \quad \text{He}_1(x) = x, \quad \text{He}_{n+1}(x) = x \text{He}_n(x) - n \text{He}_{n-1}(x).$$

Rodrigues' formula: $\text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}$. Generating function:

$$G_{\text{He}}(x, t) = \exp\left(xt - \frac{1}{2}t^2\right) = \sum_{n=0}^{\infty} \text{He}_n(x) \frac{t^n}{n!}.$$

Observation 2.3 (Singularity of Hermite). Among all classical orthogonal polynomial families, the Hermite family is the *unique* one whose orthogonality weight $w(x) = (2\pi)^{-1/2} e^{-x^2/2}$ is itself the standard Gaussian (Gauss-core) density. This isomorphism is the structural origin of Hermite's irreplaceability in kinetic-theoretic discretization.

2.4 Roots

Theorem 2.4 (Real, simple, interior roots; [21], Thm. 3.3.1). *If ϕ_n is the n -th orthogonal polynomial in $L^2(I, w)$, then ϕ_n has exactly n real, simple roots $\xi_1 < \xi_2 < \dots < \xi_n$, all of which lie in the open interior $\text{int}(I)$.*

For Legendre, $\xi_i \in (-1, 1)$; for Hermite, $\xi_i \in \mathbb{R}$ with $|\xi_i| = O(\sqrt{2n})$ as $n \rightarrow \infty$ [21, §6.32].

3. General Theory of Gauss Quadrature

3.1 Problem statement

Given a weight w on I , define the target functional

$$\mathcal{I}[f] := \int_I f(x) w(x) dx = \langle f, 1 \rangle_w, \quad (1)$$

and seek an n -point quadrature

$$\mathcal{Q}_n[f] := \sum_{i=1}^n w_i f(\xi_i) \quad (2)$$

exact for polynomials of maximal possible degree.

Remark 3.1 (Critical clarification). The functional being approximated is $\langle f, 1 \rangle_w$, not $\langle f, \phi_n \rangle_w$. The orthogonal polynomial ϕ_n enters only through the node placement $\xi_i = \text{root}(\phi_n)$ and the orthogonality-induced cancellation in the proof of precision. It does not appear in the integrand.

3.2 Precision theorem

Theorem 3.2 (Gauss precision; [6, 18]). Let $\xi_1 < \dots < \xi_n$ be the roots of ϕ_n , and define the quadrature weights via

$$w_i := \int_I \ell_i^{(n)}(x) w(x) dx, \quad \ell_i^{(n)}(x) := \prod_{j \neq i} \frac{x - \xi_j}{\xi_i - \xi_j}. \quad (3)$$

Then

$$\mathcal{Q}_n[f] = \mathcal{I}[f] \quad \text{for all } f \in \mathbb{P}_{2n-1}.$$

Moreover, no n -point rule of the form (2) is exact on all of $\mathbb{P}_{2n}$.

Proof. **Step 1 (polynomial division).** For any $f \in \mathbb{P}_{2n-1}$, the Euclidean algorithm in $\mathbb{R}[x]$ yields unique polynomials $q, r \in \mathbb{R}[x]$ with $\deg q \leq n-1$ and $\deg r \leq n-1$ such that

$$f(x) = \phi_n(x) q(x) + r(x). \quad (4)$$

Step 2 (integration against w). Multiplying (4) by $w(x)$ and integrating,

$$\mathcal{I}[f] = \int_I \phi_n(x) q(x) w(x) dx + \int_I r(x) w(x) dx = \langle \phi_n, q \rangle_w + \mathcal{I}[r]. \quad (5)$$

Step 3 (orthogonality annihilation – the key). Since $\deg q \leq n-1$ we may expand $q = \sum_{k=0}^{n-1} c_k \phi_k$. By orthogonality of $\{\phi_n\}_{n \geq 0}$,

$$\langle \phi_n, q \rangle_w = \sum_{k=0}^{n-1} c_k \underbrace{\langle \phi_n, \phi_k \rangle_w}_{=0} = 0.$$

Hence $\mathcal{I}[f] = \mathcal{I}[r]$.

Step 4 (node evaluation). Because $\phi_n(\xi_i) = 0$,

$$f(\xi_i) = \phi_n(\xi_i) q(\xi_i) + r(\xi_i) = r(\xi_i), \quad i = 1, \dots, n.$$

Step 5 (Lagrange closure). Since $r \in \mathbb{P}_{n-1}$, the n -point Lagrange interpolant of r at the nodes equals r exactly, so

$$\mathcal{I}[r] = \int_I \left[\sum_{i=1}^n r(\xi_i) \ell_i^{(n)}(x) \right] w(x) dx = \sum_{i=1}^n w_i r(\xi_i) = \sum_{i=1}^n w_i f(\xi_i) = \mathcal{Q}_n[f].$$

Combining Steps 3–5 gives $\mathcal{I}[f] = \mathcal{Q}_n[f]$.

Optimality. Take $f^*(x) := \prod_{i=1}^n (x - \xi_i)^2 \in \mathbb{P}_{2n}$. Then $f^*(\xi_i) = 0$ so $\mathcal{Q}_n[f^*] = 0$. But $f^* > 0$ on $I \setminus \{\xi_i\}$ and $w \geq 0$, whence $\mathcal{I}[f^*] > 0$. Thus $\mathcal{Q}_n[f^*] \neq \mathcal{I}[f^*]$, proving exactness on $\mathbb{P}_{2n}$ is impossible. $\square$

3.3 Closed forms for the weights

The Christoffel–Darboux identity [21, Thm. 3.2.2] yields the equivalent representations

$$w_i = \frac{h_{n-1}}{\phi_{n-1}(\xi_i) \phi_n'(\xi_i)} = - \frac{h_n}{\phi_n'(\xi_i) \phi_{n+1}(\xi_i)}, \quad (6)$$

from which the concrete formulas for Legendre and Hermite follow.

4. Gauss–Legendre Quadrature

4.1 Setup

Take $I = [-1, 1]$, $w(x) \equiv 1$, and orthogonal basis $\{P_n\}$. Theorem 3.2 yields

$$\boxed{\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i^{\text{GL}} f(\xi_i^{\text{GL}}),} \quad (7)$$

exact for $f \in \mathbb{P}_{2n-1}$.

4.2 Nodes and weights

Nodes: ξ_i^{GL} are the roots of P_n . Weights, by specialization of (6):

$$w_i^{\text{GL}} = \frac{2}{(1 - \xi_i^2) [P'_n(\xi_i)]^2}. \quad (8)$$

4.3 Tabulated values

Table 2: Gauss–Legendre nodes and weights on $[-1, 1]$ (from [1, Table 25.4]).

n	Nodes ξ_i	Weights w_i	Precision
1	0	2	1
2	± 0.5773502692	1, 1	3
3	0; ± 0.7745966692	0.8888888889; 0.5555555556	5
4	± 0.3399810436 ; ± 0.8611363116	0.6521451549; 0.3478548451	7
5	0; ± 0.5384693101 ; ± 0.9061798459	0.5688888889; 0.4786286705; 0.2369268851	9

4.4 Affine mapping to general intervals

$$\int_a^b f(x) dx = \frac{b-a}{2} \sum_{i=1}^n w_i^{\text{GL}} f\left(\frac{b-a}{2} \xi_i^{\text{GL}} + \frac{a+b}{2}\right).$$

4.5 Error estimate

For $f \in C^{2n}([-1, 1])$ [4, Eq. (2.7.12)],

$$E_n^{\text{GL}}[f] = \mathcal{I}[f] - \mathcal{Q}_n[f] = \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} f^{(2n)}(\eta), \quad \eta \in (-1, 1). \quad (9)$$

4.6 Tensor-product extension

On $[-1, 1]^D$,

$$\int_{[-1, 1]^D} f(\mathbf{x}) d^D \mathbf{x} \approx \sum_{i_1, \dots, i_D=1}^n \left(\prod_{k=1}^D w_{i_k}^{\text{GL}} \right) f(\xi_{i_1}^{\text{GL}}, \dots, \xi_{i_D}^{\text{GL}}).$$

5. Gauss–Hermite Quadrature

5.1 Setup (probabilist convention)

Take $I = \mathbb{R}$, $w(x) = (2\pi)^{-1/2}e^{-x^2/2}$, and orthogonal basis $\{\text{He}_n\}$. Theorem 3.2 yields

$$\boxed{\int_{-\infty}^{\infty} f(x) \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx \approx \sum_{i=1}^n w_i^{\text{GH}} f(\xi_i^{\text{GH}}),} \quad (10)$$

exact for $f \in \mathbb{P}_{2n-1}$ [19].

5.2 Nodes and weights

Nodes: ξ_i^{GH} are the roots of He_n . Weights:

$$w_i^{\text{GH}} = \frac{n!}{[n \text{He}_{n-1}(\xi_i)]^2}. \quad (11)$$

5.3 Tabulated values

Table 3: Gauss–Hermite nodes and weights (probabilist convention).

n	Nodes ξ_i	Weights w_i
1	0	1
2	± 1	1/2, 1/2
3	0; $\pm\sqrt{3} \approx \pm 1.7320508$	2/3; 1/6, 1/6
4	± 0.7419637 ; ± 2.3344142	0.4541242; 0.0458758
5	0; ± 1.3556262 ; ± 2.8569700	0.5333333; 0.2220759; 0.0112574

5.4 Error estimate

For $f \in C^{2n}(\mathbb{R})$ of sufficient decay,

$$E_n^{\text{GH}}[f] = \frac{n!}{(2n)!} f^{(2n)}(\eta), \quad \eta \in \mathbb{R}. \quad (12)$$

5.5 Tensor-product extension

$$\int_{\mathbb{R}^D} f(\mathbf{x}) \frac{e^{-|\mathbf{x}|^2/2}}{(2\pi)^{D/2}} d^D \mathbf{x} \approx \sum_{i_1, \dots, i_D=1}^n \left(\prod_{k=1}^D w_{i_k}^{\text{GH}} \right) f(\xi_{i_1}^{\text{GH}}, \dots, \xi_{i_D}^{\text{GH}}).$$

The separability $w_D(\mathbf{x}) = \prod_k w(x_k)$ is essential for recovering degree- $2n - 1$ exactness in each coordinate direction.

6. Comparative Analysis and Conceptual Clarifications

6.1 Three structural roles

For every Gauss- X quadrature, three distinct roles must be separated; see Table 4.

Table 4: Three structural roles in Gauss- X quadrature.

Role	Content	Notation
Target functional	integral to be evaluated	$\mathcal{I}[f] = \langle f, 1 \rangle_w$
Quadrature nodes	sample locations	$\xi_i = \text{root}(\phi_n)$
Annihilation mechanism	enables $2n - 1$ exactness	$\langle \phi_n, q \rangle_w = 0, \deg q < n$

6.2 Completeness interval = quadrature range

Theorem 6.1 (Completeness–quadrature lock). *For Gauss- X quadrature derived from the family $\{\phi_n\}$ orthogonal on (I, w) , the natural domain of integration coincides with I .*

Proof sketch. (i) $\text{supp}(w) \subseteq I$ defines the nontrivial portion of $\mathcal{I}[f]$. (ii) By Theorem 2.4, all roots of ϕ_n lie in $\text{int}(I)$. (iii) Hence $\mathcal{Q}_n[f]$ samples f only in $\text{int}(I)$. The three are mutually locked to I . $\square$

6.3 Selection rule

Table 5: Selection rule for Gauss- X quadrature based on the physical setting.

Physical setting	Appropriate quadrature	Reason
Bounded interval, unit weight	Gauss–Legendre	$w \equiv 1$ on $[-1, 1]$
Unbounded interval, Gaussian decay	Gauss–Hermite	$w = \text{Gauss-core density}$
Half-line, exponential decay	Gauss–Laguerre	$w(x) = e^{-x}$ on $[0, \infty)$

7. The Hermite–Maxwell Isomorphism and Velocity-Space Discretization

7.1 Statement

Theorem 7.1 (Kinetic-theoretic uniqueness of the Hermite weight; [16, 17]). *Among the classical orthogonal polynomial families, the Hermite family is the unique one whose orthogonality weight is isomorphic, under a linear rescaling of the integration variable, to the local Maxwell–Boltzmann equilibrium distribution.*

Explicit form. The isothermal Maxwell distribution

$$f^{\text{MB}}(\boldsymbol{\xi}; \rho, \mathbf{u}, T) = \frac{\rho}{(2\pi RT)^{D/2}} \exp\left(-\frac{|\boldsymbol{\xi} - \mathbf{u}|^2}{2RT}\right)$$

admits, on setting $c_s^2 := RT$ and $\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}/c_s$,

$$f^{\text{MB}}(\boldsymbol{\xi}) c_s^D \propto \rho w(\tilde{\boldsymbol{\xi}} - \mathbf{u}/c_s), \quad w(\mathbf{y}) = (2\pi)^{-D/2} e^{-|\mathbf{y}|^2/2}.$$

7.2 Implication for LBM velocity discretization

Velocity moments take the form

$$\mathcal{M}_{\alpha_1 \dots \alpha_k}^{(k)} = \int_{\mathbb{R}^D} \xi_{\alpha_1} \dots \xi_{\alpha_k} f^{\text{eq}}(\boldsymbol{\xi}) d^D \boldsymbol{\xi}.$$

After Hermite expansion of f^{eq}/w , the integrand becomes a polynomial in $\boldsymbol{\xi}$ multiplied by the Gauss-core weight; by Theorem 3.2 an n -point Gauss–Hermite rule applied coordinate-wise exactly preserves any moment whose associated polynomial has degree $\leq 2n - 1$.

Corollary 7.2 (Origin of D2Q9, D3Q19, D3Q27). *The three-point one-dimensional Gauss–Hermite rule has nodes $\tilde{\xi}_i \in \{0, \pm\sqrt{3}\}$. The choice $c_s = 1/\sqrt{3}$ rescales these onto the integer lattice $\xi_i \in \{0, \pm 1\}$. The D -fold tensor product yields D2Q9, D3Q19, and D3Q27 (the latter two after weight-pruning of dynamically negligible diagonal nodes) [7, 14].*

7.3 Truncated equilibrium

Truncating at $N = 2$ recovers the standard Navier–Stokes-level equilibrium,

$$f_\alpha^{\text{eq}} = w_\alpha \rho \left[1 + \frac{\boldsymbol{\xi}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{(\boldsymbol{\xi}_\alpha \cdot \mathbf{u})^2}{2c_s^4} - \frac{|\mathbf{u}|^2}{2c_s^2} \right]. \quad (13)$$

8. Gauss–Legendre in GILBM Physical-Space Interpolation

8.1 Interpolation problem

In GILBM the streaming step may require f_α^* at off-lattice positions $\mathbf{x}^*$. Given lattice values $\{f_\alpha^*(\mathbf{x}_{i,j,k}, t)\}$, the reconstruction

$$f_\alpha^*(\mathbf{x}^*, t) = \sum_{i,j,k} N_{i,j,k}(\mathbf{x}^*) f_\alpha^*(\mathbf{x}_{i,j,k}, t)$$

requires basis functions $N_{i,j,k}$ defined on a local cell.

8.2 Optimality of Legendre centers

Normalize the local cell to $[-1, 1]^D$. The natural inner product is

$$\langle f, g \rangle = \int_{[-1, 1]^D} f(\mathbf{x}) g(\mathbf{x}) d^D \mathbf{x},$$

with unit weight $w \equiv 1$.

Theorem 8.1 (Optimality of Legendre interpolation centers). *Let $\{\xi_i^{\text{GL}}\}_{i=1}^n$ be the Gauss–Legendre nodes on $[-1, 1]$. The tensor-product Lagrange interpolant at $\{\xi_{i_1}^{\text{GL}}, \dots, \xi_{i_D}^{\text{GL}}\}$ exactly integrates the polynomial space $\mathbb{P}_{2n-1}^{\otimes D}$ on $[-1, 1]^D$, and is the optimal n^D -point interpolation stencil for unit-weight integrals on the local cell.*

8.3 Non-interchangeability

Table 6: Non-interchangeable assignment of quadrature schemes in GILBM.

Sub-problem	Completeness interval	Required quadrature
Velocity-moment evaluation	$\mathbb{R}^D$ (Hermite)	Gauss–Hermite
Physical-cell interpolation	$[-1, 1]^D$ (Legendre)	Gauss–Legendre

By Theorem 6.1 the assignments cannot be swapped: Gauss–Hermite nodes spread on the scale $\sqrt{2n}$, bearing no relation to a bounded cell; Gauss–Legendre nodes are confined to $(-1, 1)$ and cannot capture the Maxwellian tail.

9. Discussion and Outlook

We have given a unified, self-contained derivation of the Gauss–Hermite and Gauss–Legendre quadrature rules in the weighted- L^2 framework, and shown how the two rules occupy complementary, non-interchangeable roles in the GILBM. Three conceptual points warrant emphasis:

1. The target functional is $\langle f, 1 \rangle_w$, not $\langle f, \phi_n \rangle_w$. The orthogonal polynomial enters only as a node generator and as the agent of orthogonality cancellation.
2. The completeness interval is the quadrature range (Theorem 6.1); this is a theorem, not a convention.
3. The Hermite weight is the standard Gaussian (Observation 2.3); this isomorphism with the Maxwell–Boltzmann distribution is *unique* among the classical families and underpins the canonical kinetic-discretization tool.

For higher-order GILBM development the framework suggests: (i) higher-order Hermite expansions ($N \geq 3$) to capture compressibility and thermal effects [17, 5]; (ii) higher-order Gauss–Legendre stencils on the physical-space cell, especially near solid boundaries; (iii) coupling of the two quadrature schemes on non-uniform grids, possibly via Gauss–Lobatto deformation at boundaries [10].

A. Conversion between Hermite conventions

The probabilist and physicist Hermite polynomials are related by

$$H_n(x) = 2^{n/2} \text{He}_n(\sqrt{2}x), \quad \text{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2}).$$

The quadrature nodes and weights transform as $\xi_i^{(H)} = \xi_i^{(\text{He})}/\sqrt{2}$ and $w_i^{(H)} = w_i^{(\text{He})} \cdot \sqrt{\pi}$.

B. Unified view via Gauss–Jacobi

All classical Gauss rules are specializations of the Gauss–Jacobi family with weight

$$w_{\alpha,\beta}(x) = (1-x)^\alpha(1+x)^\beta, \quad \alpha, \beta > -1, \quad x \in [-1, 1].$$

Legendre corresponds to $\alpha = \beta = 0$; continuous deformation in (α, β) interpolates among Chebyshev, Legendre and other Jacobi variants, providing a unified perspective for non-standard weighted LBM models [21, Ch. 4].

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